COMPLETION ALONG A CLOSED SUBSET IS COMPATIBLE WITH BASE CHANGE OF ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted spans are self-contained: no reference leaves them and no bibliography citation occurs in them.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The completion of an algebraic space along a closed subset, as a sheaf on the fppf site. Same chapter, Definition definition-completion and the construction of X_{/T}. Supplies the objects occurring in the square and the meaning of the cartesian property.
- Morphisms representable by algebraic spaces, tested on schemes. Bootstrap, the sections on morphisms representable by spaces. Gives the meaning of the final 'in particular' clause and lets the cartesian property be tested on Y -> X.

Source reference. Original lemma 0AIZ: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 3179–3179. Pointer outside the counted span, invoked only from a delivered context span as a forward reference; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Definition 0.1. Let S be a scheme. Let X be an algebraic space over S. Let $T \subset |X|$ be a closed subset. The formal algebraic space of Lemma 0AIZ is called the *completion of X along T*.

Lemma 0.2. *Let S be a scheme. Let $f : X' \to X$ be a morphism of algebraic spaces over S. Let $T \subset |X|$ be a closed subset and let $T' = |f|^{-1}(T) \subset |X'|$. Then*

$$\begin{array}{ccc} X'_{/T'} & \longrightarrow & X' \\ \downarrow & & \downarrow f \\ X_{/T} & \longrightarrow & X \end{array}$$

is a cartesian diagram of sheaves. In particular, the morphism $X'_{/T'} \to X_{/T}$ is representable by algebraic spaces.

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Proof. Namely, suppose that $Y \to X$ is a morphism from a scheme into X such that $|Y|$ maps into T. Then $Y \times_X X' \to X$ is a morphism of algebraic spaces such that $|Y \times_X X'|$ maps into T'. Hence the functor $Y \times_{X_{/T}} X'_{/T'}$ is represented by $Y \times_X X'$ and we see that the lemma holds. $\square$